VIVA LA CRESTA · OPTIMISER REPORT

The fastest way down

What the game's own physics allows from Junction and from Top: the best plan found for when to steer, when to sit back on the knives and when to rake, for every rider on both toboggans, and the best ways to recover from a bad entry into Shuttlecock.

38.27 s	49.92 s	5.13 s	0.92 M
Fastest from Junction (The Dark Lord, flat-top)	Fastest from Top (The Dark Lord, flat-top)	Gained on the autopilot's good flat-top ride from Junction (Adelaide)	Rides simulated, in 22 minutes on a laptop

KEY FINDINGS

- **The autopilot leaves 4.2–8.7 s on the run** in every case. The optimum's time comes from braking far less: Adelaide from Junction on the flat-top rakes the equivalent of 15 m at full rake, against 49 m for the autopilot, and goes back on the knives only briefly before and through Shuttlecock.
- **Most of it is won after Shuttlecock**: the speed carried out of Shuttlecock and Stream is worth the most on Bledisloe Straight, the longest stretch of the run.
- **Every Shuttlecock mistake tried can be saved** (8 of 8 on the flat-top, 8 of 8 on the traditional), most at no cost at all, extra speed included: rake early, from the moment you see it. Only arriving too slow costs real time (0.75 s on the flat-top, 0.28 s on the traditional). But riding on as if nothing had happened falls in 9 of the 16 cases: the remedy is what matters.
- **Fastest riders** from Junction on the flat-top: The Dark Lord 38.27 s, Maximilian 40.10 s, Mario 40.13 s; slowest Kletus (41.73 s). The flat-top is worth about 2.5 s from Junction.

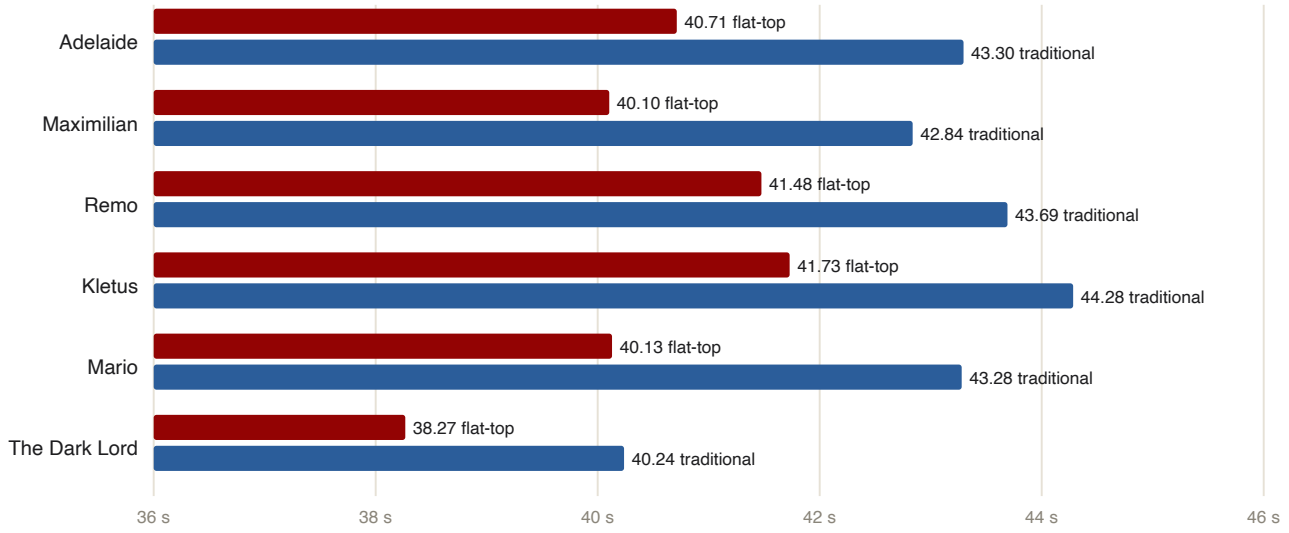
HOW IT WORKS

- **A plan along the run.** Every 10 m of track the plan sets four things: how high to ride the banks (a share of the balanced line for the speed), whether to be back on the knives, the speed above which to rake with both feet, and a steer of its own. The rider steers onto the line by feedback, as a person does, and adds the plan's own steer, so it can take lines the feedback alone never would. From Junction that is about 360 numbers, from Top about 490. The running start strides every 0.15 s, about as quick as a thumb taps.
- **Scored by riding it.** Each plan is ridden in the game's physics (the same code as the game, 240 steps a second, no randomness): the score is the time from the start beam to the finish, and a fall scores far worse, less so the further down it happened.
- **Searched with CMA-ES**, an evolution strategy: generations of plans sampled around the best so far, the search narrowing as it learns which numbers matter. It starts from the autopilot's choreography, then a last sweep nudges each number in turn and keeps what helps. About 470 rides a second on nine worker threads.
- **Then Shuttlecock.** From the optimum's own state 17 m before Shuttlecock, eight different mistakes (too fast, too slow, too high, too low, turned out, late onto the knives), each optimised again to the finish.

Every result is stored in `optimiser/results/` (one JSON file per case: the plan, the ride's trace and the comparison), so the plans can be used again: a ghost to race, the ideal line on the My times map, or a check on the club's scratch times.

All riders from Junction

BEST TIMES (S)

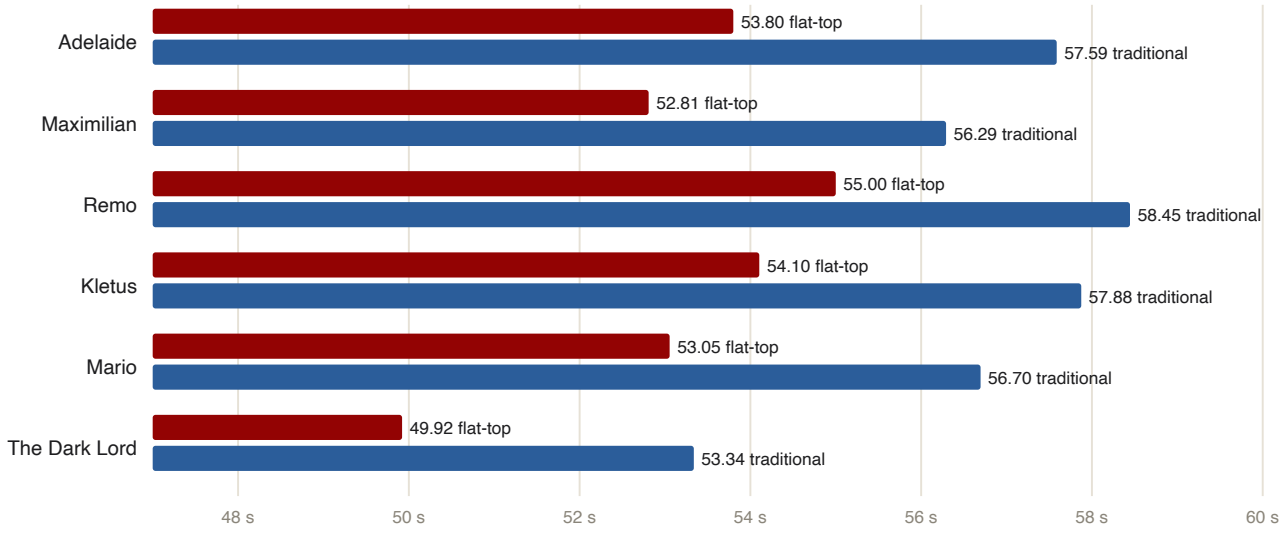


OPTIMUM AGAINST THE AUTOPILOT (S)

Rider	Flat-top			Traditional		
	Optimum	**Autopilot**	**Gain**	**Optimum**	**Autopilot**	**Gain**
Adelaide	**40.71**	45.84	–5.13	**43.30**	49.77	–6.48
Maximilian	**40.10**	45.63	–5.53	**42.84**	49.64	–6.80
Remo	**41.48**	45.99	–4.52	**43.69**	49.87	–6.18
Kletus	**41.73**	45.90	–4.18	**44.28**	49.87	–5.59
Mario	**40.13**	45.26	–5.13	**43.28**	49.22	–5.94
The Dark Lord	**38.27**	43.89	–5.63	**40.24**	47.81	–7.58

All riders from Top

BEST TIMES (S)

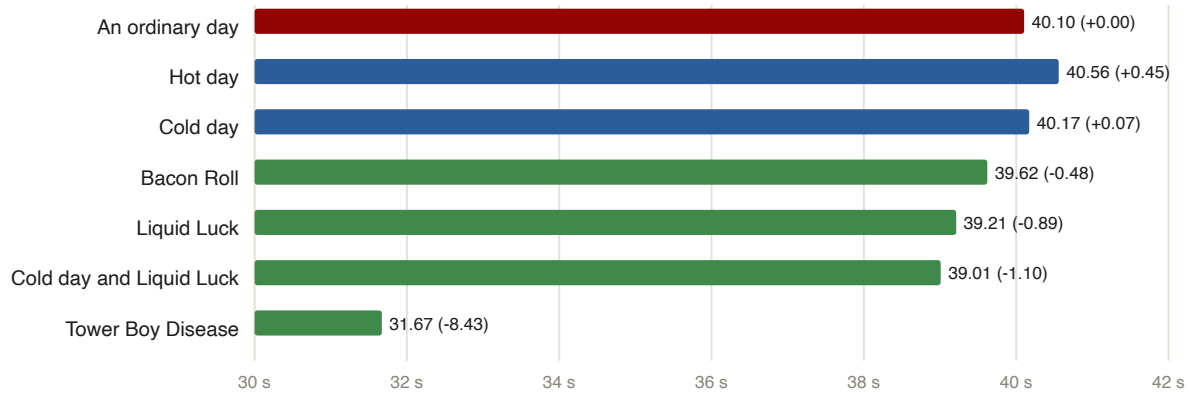


OPTIMUM AGAINST THE AUTOPILOT (S)

Rider	Flat-top			Traditional		
	Optimum	**Autopilot**	**Gain**	**Optimum**	**Autopilot**	**Gain**
Adelaide	**53.80**	60.55	–6.75	**57.59**	65.12	–7.53
Maximilian	**52.81**	60.31	–7.50	**56.29**	64.91	–8.62
Remo	**55.00**	fell (Shuttlecock)	—	**58.45**	65.28	–6.83
Kletus	**54.10**	60.94	–6.83	**57.88**	65.25	–7.38
Mario	**53.05**	60.16	–7.10	**56.70**	64.55	–7.86
The Dark Lord	**49.92**	57.96	–8.04	**53.34**	62.05	–8.71

What the day is worth

Maximilian from Junction on the flat-top, the plan optimised again for each kind of day the game can deal (the weather, and the luck at the start). On an ordinary day the best found is **40.10 s**.



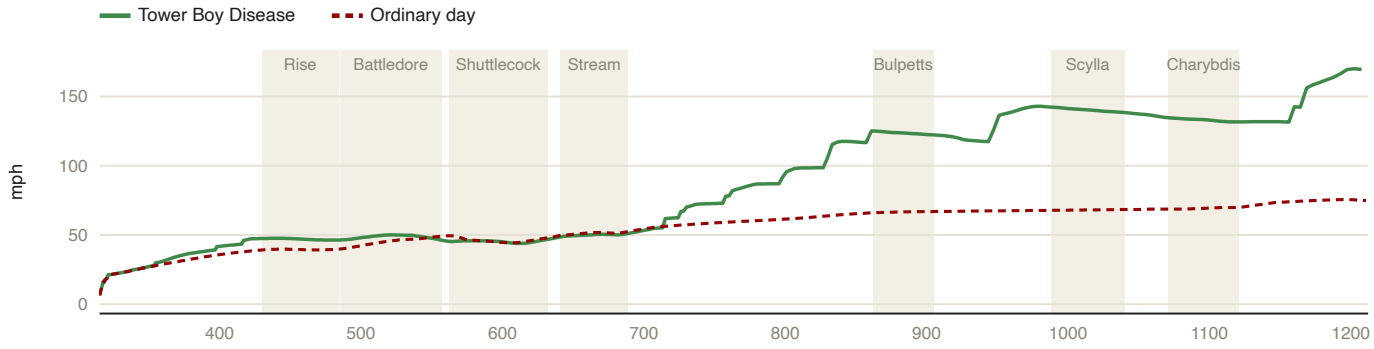
A 39.6 FROM JUNCTION

Faster than the optimum for an ordinary day by 0.50 s. It is within reach on a day with Liquid Luck (best 39.21 s), nearly with a Bacon Roll (39.62 s), and easily with Tower Boy Disease. Without any of them it would mean a line the optimiser hasn't found: the search starts from the autopilot's way of riding and refines it, and a quite different line could beat it. A search with much bigger steps, to look for quite different lines, didn't find a faster one either.

TWO THINGS THE GAME MIGHT WANT TO LOOK AT

- **The Bacon Roll is a help, not a handicap**: 0.48 s faster. Thirty per cent more weight carries more speed against the air; the rakes bite less on a heavier rider, but the optimum hardly rakes.
- **Tower Boy Disease is worth 8.43 seconds** (31.67 s): the optimum hits the bankinas on purpose, 20 times, each knock a boost.

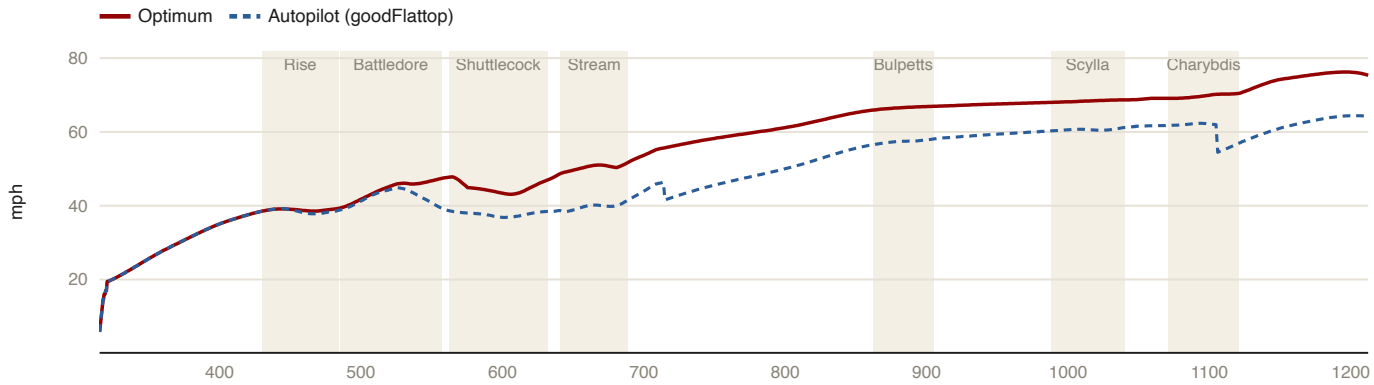
SPEED WITH TOWER BOY DISEASE, AGAINST AN ORDINARY DAY



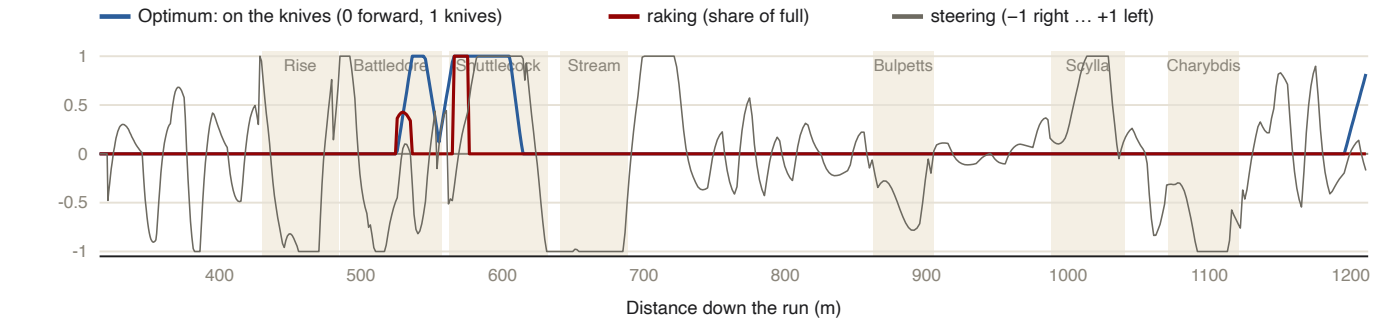
From Junction on the flat-top

Adelaide on the flat-top: optimum **40.71 s**, against 45.84 s for the autopilot's choreography (“goodFlattop”) — **5.13 s** faster. Top speed 76 mph, 75 mph through the finish; 0 bankina knocks.

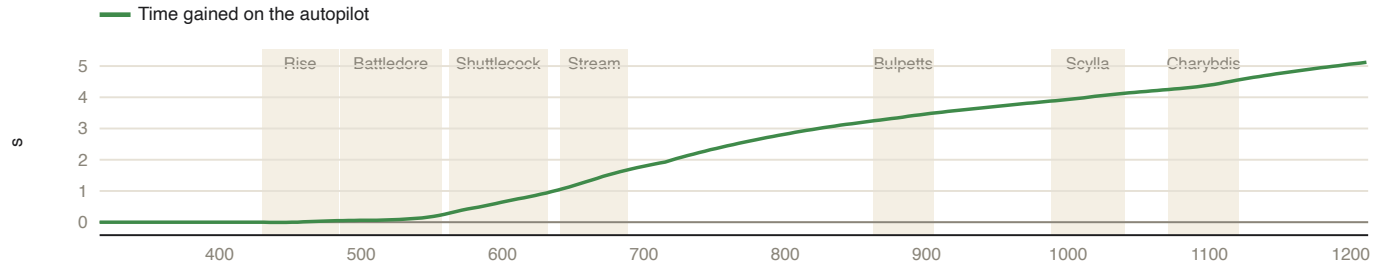
SPEED DOWN THE RUN



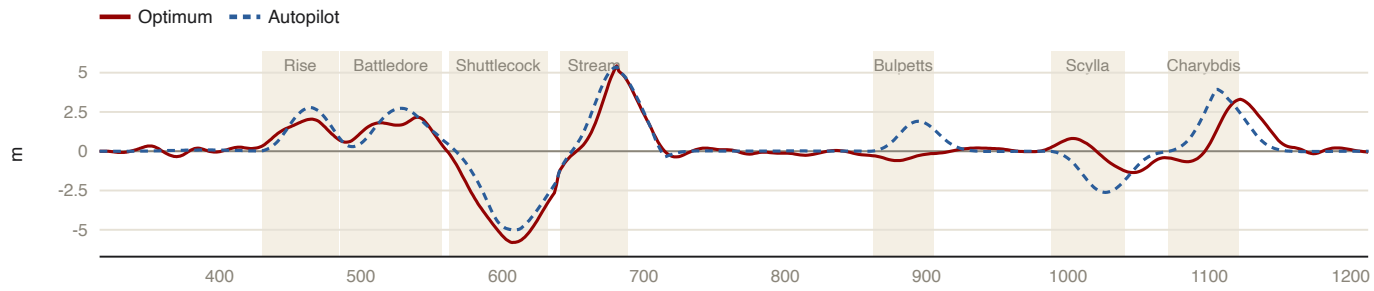
WHAT THE OPTIMUM DOES



WHERE THE TIME IS WON (SECONDS AHEAD OF THE AUTOPILOT)



LINE ON THE ICE (+ LEFT, – RIGHT; THE FLOOR'S EDGES ARE AT ABOUT ±0.65 M)

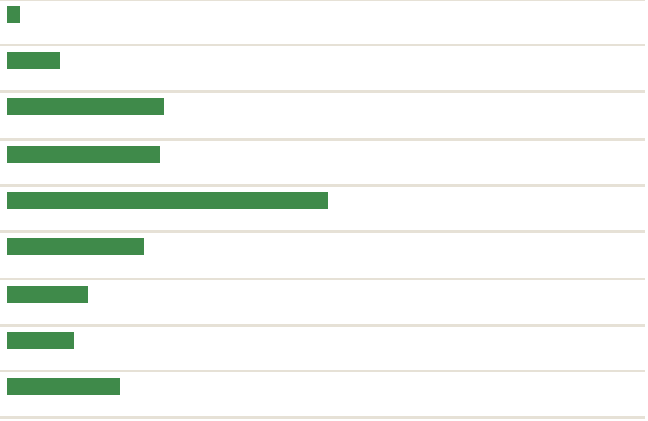


	Rise	Stream	Bulpetts	Finish
Optimum	11.38	22.85	29.59	**40.71**
Autopilot	11.41	24.54	32.86	45.84

From Junction on the flat-top: the plan in words

Time won on the autopilot, section by section

Section	From	Won
Junction Straight	315 m	-0.00 s
Rise	426 m	+0.05 s
Battledore	485 m	+0.25 s
Shuttlecock	560 m	+0.73 s
Stream	638 m	+0.71 s
Bledisloe Straight	692 m	+1.51 s
Bulpetts	862 m	+0.64 s
Scylla	990 m	+0.37 s
Charybdis	1072 m	+0.31 s
Cresta Leap	1120 m	+0.52 s



When to sit back and rake

Where	Section	Action	Speed
526–535 m	Battledore	Rake (both feet, up to 43%)	46 → 46 mph
531–550 m	Battledore	Back on the knives	46 → 47 mph
560–609 m	Shuttlecock	Back on the knives	48 → 43 mph
566–575 m	Shuttlecock	Rake (both feet, up to 100%)	48 → 45 mph

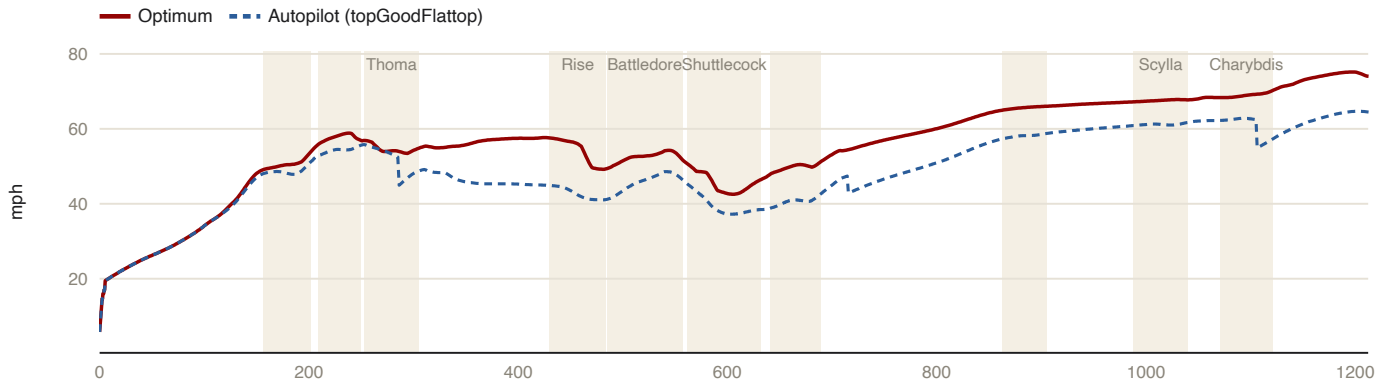
Line through the bends

Bend	Line (share of balanced)	Highest on the ice	Top speed	On the knives
Rise	60%	2.04 m	39 mph	0%
Battledore	87%	2.17 m	47 mph	27%
Shuttlecock	85%	5.81 m	48 mph	69%
Stream	118%	5.44 m	52 mph	0%
Bulpetts	112%	0.60 m	67 mph	0%
Scylla	95%	1.22 m	69 mph	0%
Charybdis	96%	3.28 m	70 mph	0%

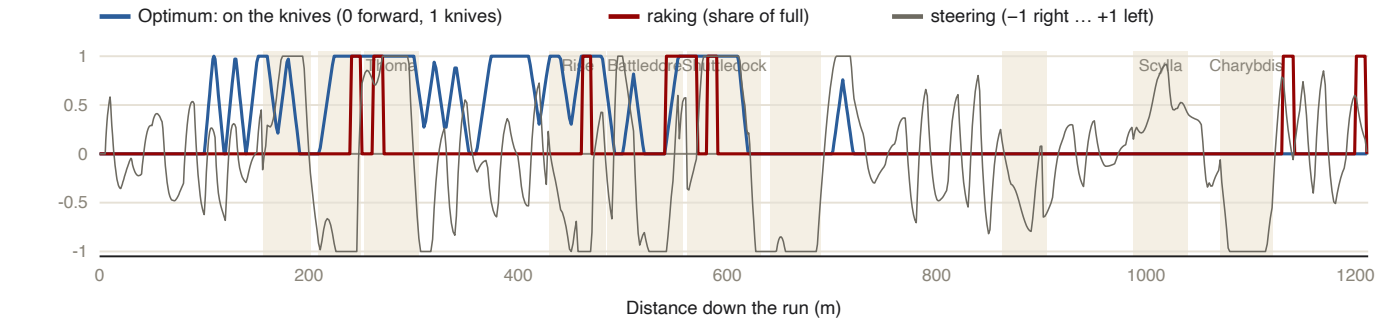
From Top on the flat-top

Adelaide on the flat-top: optimum **53.80 s**, against 60.55 s for the autopilot's choreography (“topGoodFlattop”) — **6.75 s** faster. Top speed 75 mph, 74 mph through the finish; 0 bankina knocks.

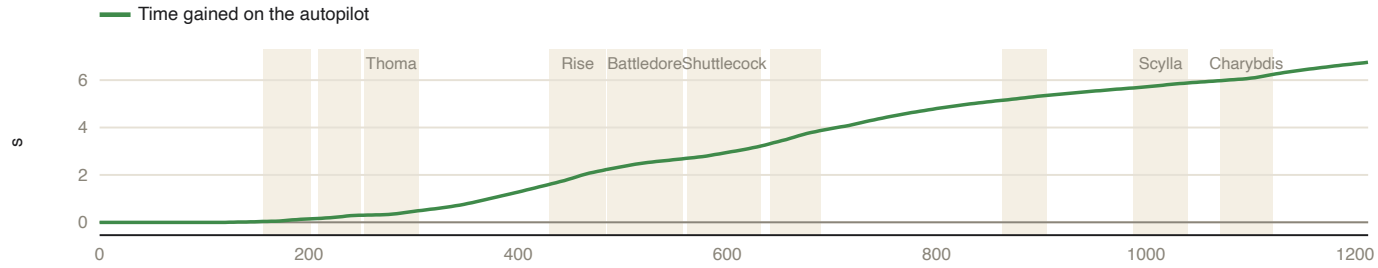
SPEED DOWN THE RUN



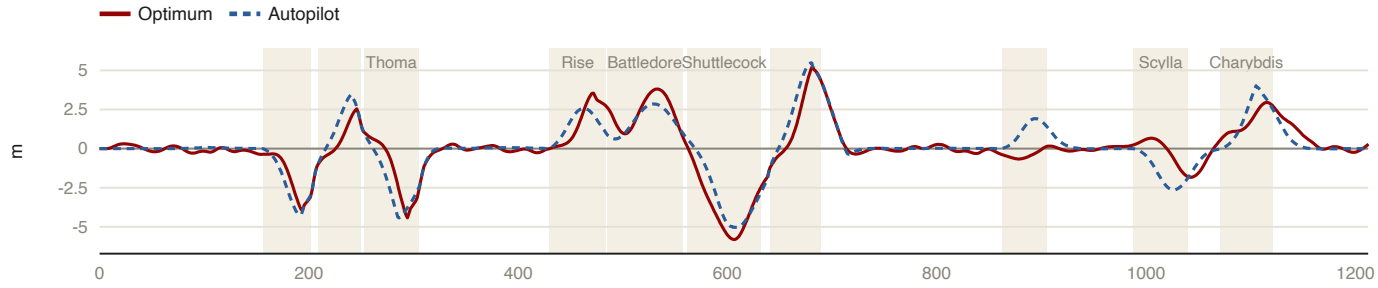
WHAT THE OPTIMUM DOES



WHERE THE TIME IS WON (SECONDS AHEAD OF THE AUTOPILOT)



LINE ON THE ICE (+ LEFT, – RIGHT; THE FLOOR'S EDGES ARE AT ABOUT ±0.65 M)

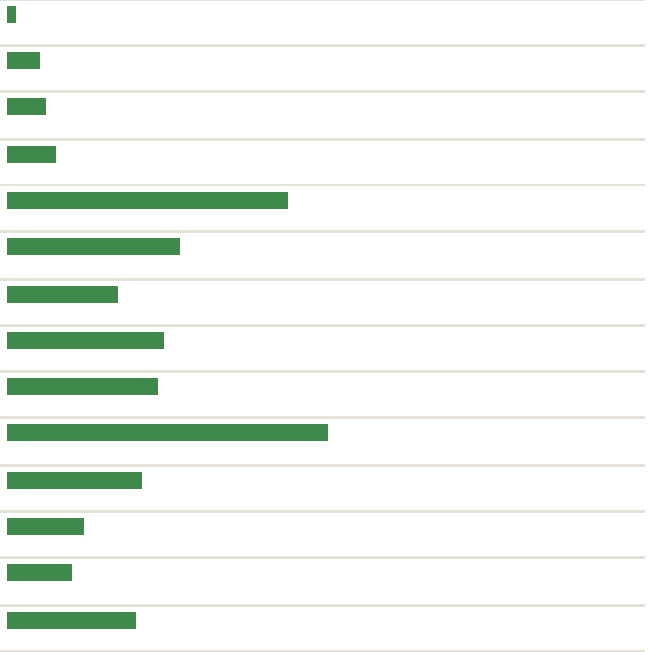


	Junction	Rise	Stream	Bulpetts	Finish
Optimum	18.78	24.94	35.64	42.51	**53.80**
Autopilot	19.34	27.03	39.53	47.69	60.55

From Top on the flat-top: the plan in words

Time won on the autopilot, section by section

Section	From	Won
Top	0 m	-0.00 s
Church Leap	110 m	+0.03 s
Curzon	150 m	+0.12 s
Brabazon	202 m	+0.15 s
Thoma	250 m	+0.19 s
Junction Straight	302 m	+1.10 s
Rise	426 m	+0.68 s
Battledore	485 m	+0.44 s
Shuttlecock	560 m	+0.62 s
Stream	638 m	+0.59 s
Bledisloe Straight	692 m	+1.26 s
Bulpetts	862 m	+0.53 s
Scylla	990 m	+0.30 s
Charybdis	1072 m	+0.25 s
Cresta Leap	1120 m	+0.50 s



When to sit back and rake

Where	Section	Action	Speed
105–114 m	Top → Church Leap	Back on the knives	35 → 37 mph
126–134 m	Church Leap	Back on the knives	40 → 42 mph
146–166 m	Church Leap → Curzon	Back on the knives	47 → 50 mph
175–185 m	Curzon	Back on the knives	50 → 51 mph
217–307 m	Brabazon → Junction Straight	Back on the knives	57 → 55 mph
241–249 m	Brabazon	Rake (both feet, up to 100%)	59 → 57 mph
262–270 m	Thoma	Rake (both feet, up to 100%)	56 → 54 mph
315–326 m	Junction Straight	Back on the knives	55 → 55 mph
334–346 m	Junction Straight	Back on the knives	55 → 56 mph
368–417 m	Junction Straight	Back on the knives	57 → 57 mph
424–446 m	Junction Straight → Rise	Back on the knives	58 → 57 mph
454–485 m	Rise → Battledore	Back on the knives	56 → 49 mph
462–469 m	Rise	Rake (both feet, up to 100%)	54 → 50 mph

Line through the bends

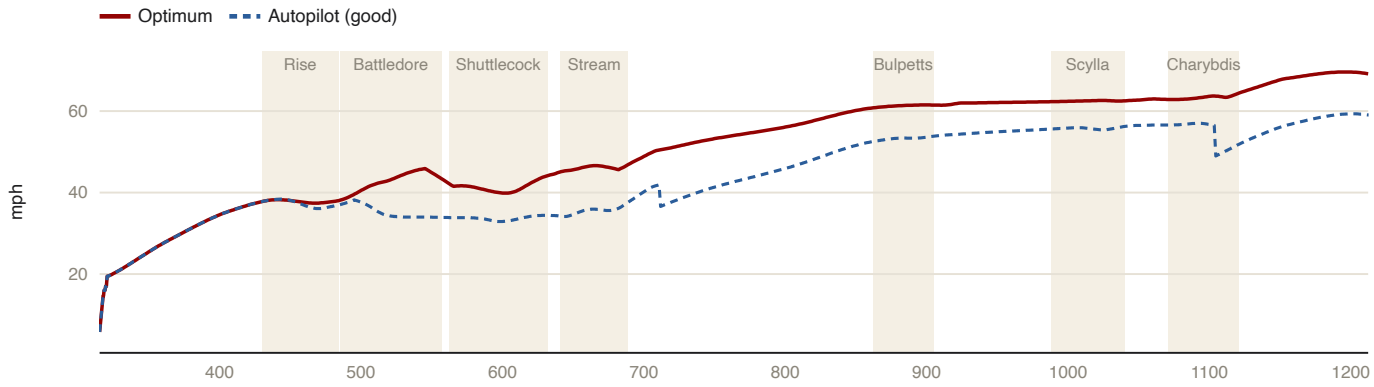
Bend	Line (share of balanced)	Highest on the ice	Top speed	On the knives
Curzon	78%	3.93 m	54 mph	47%
Brabazon	105%	2.54 m	59 mph	80%
Thoma	85%	4.43 m	57 mph	100%
Rise	94%	3.55 m	58 mph	89%
Battledore	60%	3.81 m	54 mph	25%
Shuttlecock	88%	5.81 m	50 mph	71%
Stream	110%	5.20 m	51 mph	0%
Bulpetts	121%	0.66 m	66 mph	0%
Scylla	97%	1.77 m	68 mph	0%
Charybdis	66%	2.95 m	70 mph	0%

542–570 m	Battledore → Shuttlecock	Rake (both feet, up to 100%)	54 → 49 mph
548–614 m	Battledore → Shuttlecock	Back on the knives	54 → 43 mph
581–590 m	Shuttlecock	Rake (both feet, up to 100%)	48 → 44 mph
1131–1141 m	Cresta Leap	Rake (both feet, up to 100%)	71 → 72 mph
1201–1209 m	Cresta Leap	Rake (both feet, up to 100%)	75 → 74 mph

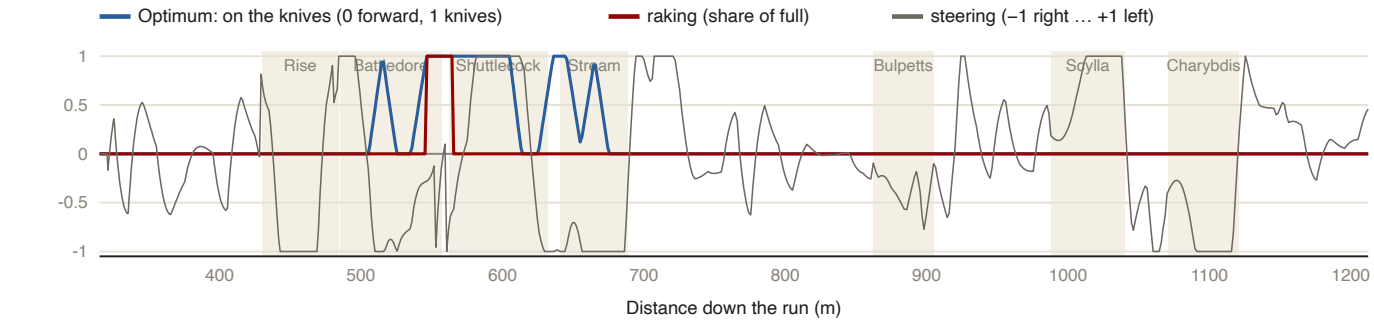
From Junction on the traditional

Adelaide on the traditional: optimum **43.30 s**, against 49.77 s for the autopilot's choreography (“good”) — **6.48 s** faster. Top speed 70 mph, 69 mph through the finish; 0 bankina knocks.

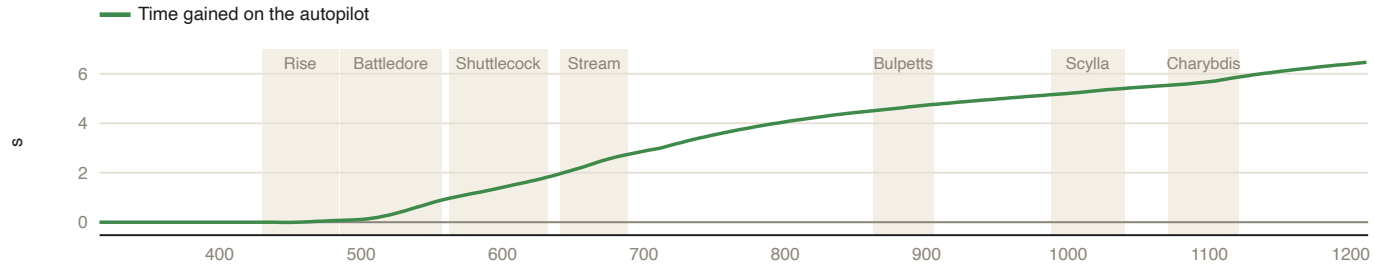
SPEED DOWN THE RUN



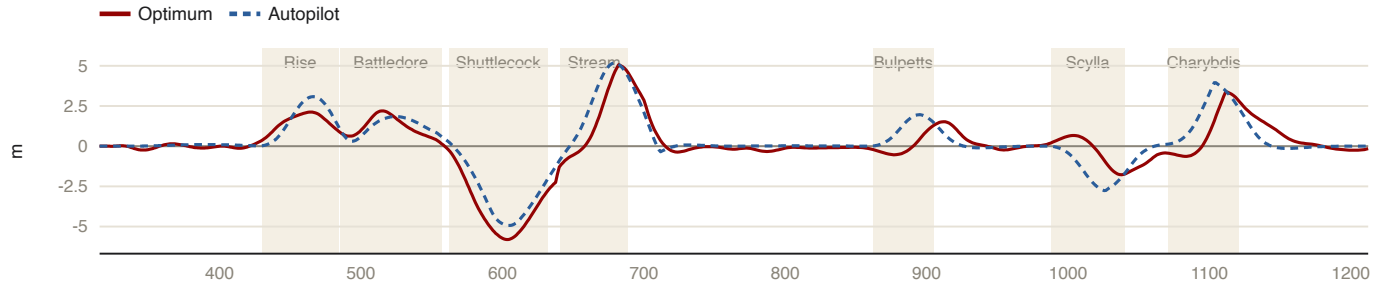
WHAT THE OPTIMUM DOES



WHERE THE TIME IS WON (SECONDS AHEAD OF THE AUTOPILOT)



LINE ON THE ICE (+ LEFT, − RIGHT; THE FLOOR'S EDGES ARE AT ABOUT ±0.65 M)

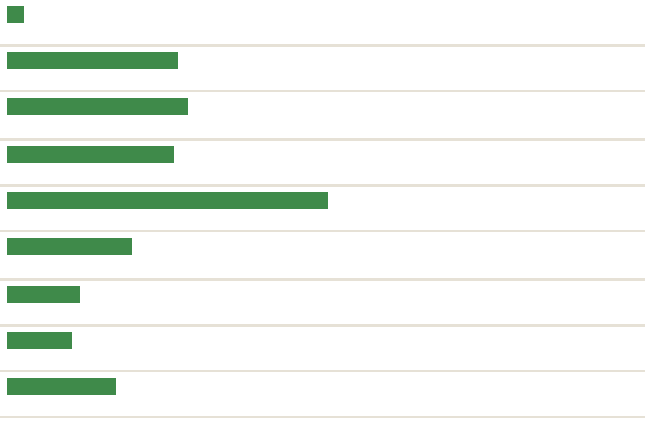


	Rise	Stream	Bulpetts	Finish
Optimum	11.53	23.75	31.11	**43.30**
Autopilot	11.57	26.51	35.65	49.77

From Junction on the traditional: the plan in words

Time won on the autopilot, section by section

Section	From	Won
Junction Straight	315 m	-0.00 s
Rise	426 m	+0.08 s
Battledore	485 m	+0.90 s
Shuttlecock	560 m	+0.96 s
Stream	638 m	+0.88 s
Bledisloe Straight	692 m	+1.70 s
Bulpetts	862 m	+0.65 s
Scylla	990 m	+0.38 s
Charybdis	1072 m	+0.34 s
Cresta Leap	1120 m	+0.58 s



When to sit back and rake

Where	Section	Action	Speed
511–519 m	Battledore	Back on the knives	42 → 43 mph
541–609 m	Battledore → Shuttlecock	Back on the knives	46 → 40 mph
547–564 m	Battledore → Shuttlecock	Rake (both feet, up to 100%)	46 → 42 mph
631–649 m	Shuttlecock → Stream	Back on the knives	44 → 46 mph
660–670 m	Stream	Back on the knives	46 → 47 mph

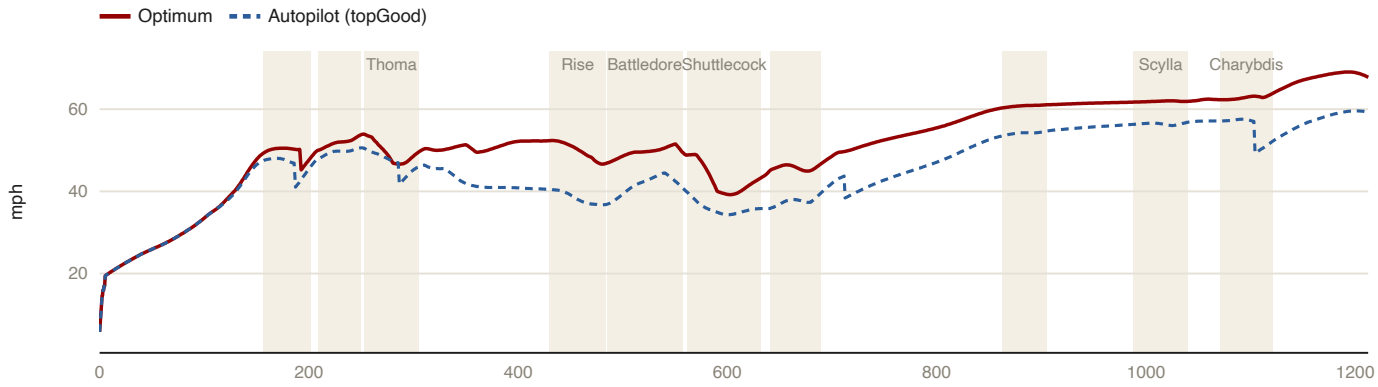
Line through the bends

Bend	Line (share of balanced)	Highest on the ice	Top speed	On the knives
Rise	74%	2.12 m	38 mph	0%
Battledore	78%	2.20 m	46 mph	36%
Shuttlecock	88%	5.81 m	44 mph	71%
Stream	121%	5.07 m	47 mph	41%
Bulpetts	45%	1.29 m	62 mph	0%
Scylla	72%	1.77 m	63 mph	0%
Charybdis	108%	3.37 m	64 mph	0%

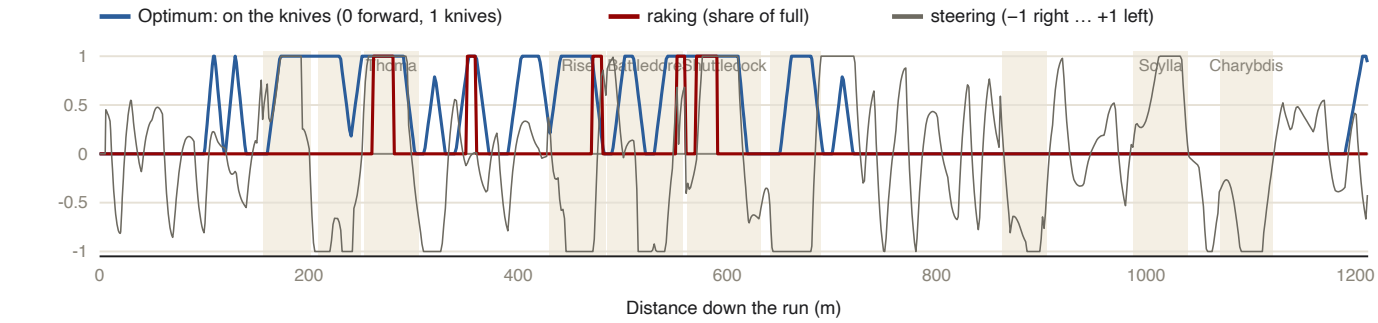
From Top on the traditional

Adelaide on the traditional: optimum **57.59 s**, against 65.12 s for the autopilot's choreography (“topGood”) — **7.53 s** faster. Top speed 69 mph, 68 mph through the finish; 1 bankina knock.

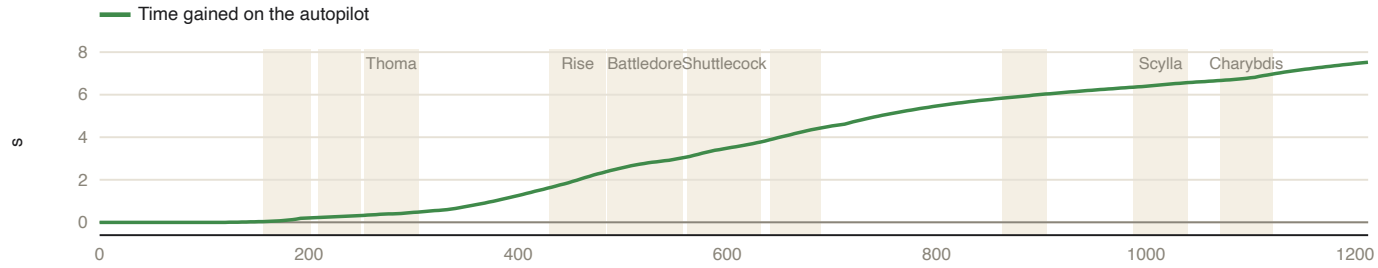
SPEED DOWN THE RUN



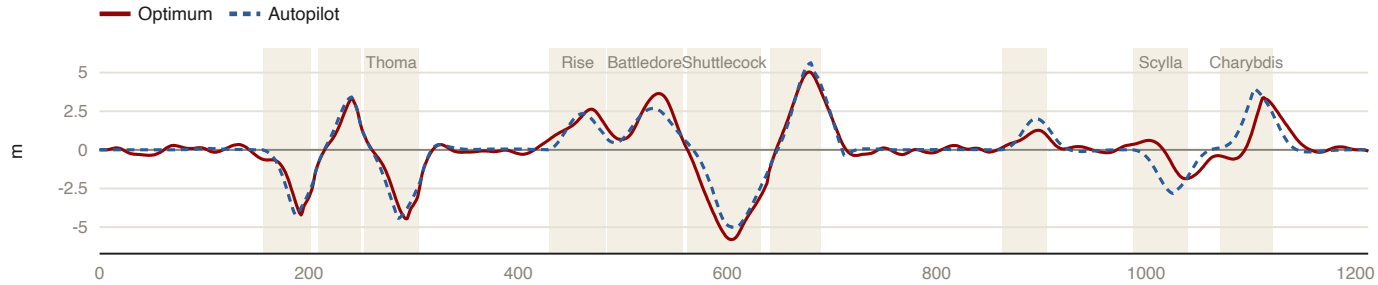
WHAT THE OPTIMUM DOES



WHERE THE TIME IS WON (SECONDS AHEAD OF THE AUTOPILOT)



LINE ON THE ICE (+ LEFT, − RIGHT; THE FLOOR'S EDGES ARE AT ABOUT ±0.65 M)



	Junction	Rise	Stream	Bulpetts	Finish
Optimum	19.56	26.37	37.84	45.28	**57.59**
Autopilot	20.10	28.56	42.28	51.14	65.12

From Top on the traditional: the plan in words

Time won on the autopilot, section by section

Section	From	Won
Top	0 m	-0.00 s
Church Leap	110 m	+0.03 s
Curzon	150 m	+0.19 s
Brabazon	202 m	+0.11 s
Thoma	250 m	+0.15 s
Junction Straight	302 m	+1.12 s
Rise	426 m	+0.82 s
Battledore	485 m	+0.63 s
Shuttlecock	560 m	+0.82 s
Stream	638 m	+0.59 s
Bledisloe Straight	692 m	+1.38 s
Bulpetts	862 m	+0.53 s
Scylla	990 m	+0.30 s
Charybdis	1072 m	+0.29 s
Cresta Leap	1120 m	+0.54 s

When to sit back and rake

Where	Section	Action	Speed
105–114 m	Top → Church Leap	Back on the knives	34 → 36 mph
125–134 m	Church Leap	Back on the knives	39 → 42 mph
168–235 m	Curzon → Brabazon	Back on the knives	50 → 52 mph
245–295 m	Brabazon → Thoma	Back on the knives	53 → 48 mph
262–280 m	Thoma	Rake (both feet, up to 100%)	53 → 47 mph
347–365 m	Junction Straight	Back on the knives	51 → 50 mph
351–359 m	Junction Straight	Rake (both feet, up to 100%)	51 → 50 mph
398–426 m	Junction Straight	Back on the knives	52 → 52 mph
435–476 m	Rise	Back on the knives	52 → 47 mph
471–480 m	Rise	Rake (both feet, up to 100%)	48 → 47 mph
496–515 m	Battledore	Back on the knives	48 → 50 mph
537–614 m	Battledore → Shuttlecock	Back on the knives	50 → 40 mph
552–559 m	Battledore	Rake (both feet, up to 100%)	51 → 49 mph

Line through the bends

Bend	Line (share of balanced)	Highest on the ice	Top speed	On the knives
Curzon	114%	4.21 m	51 mph	79%
Brabazon	95%	3.34 m	54 mph	79%
Thoma	64%	4.46 m	54 mph	84%
Rise	55%	2.64 m	52 mph	76%
Battledore	97%	3.65 m	52 mph	55%
Shuttlecock	85%	5.80 m	49 mph	74%
Stream	50%	5.04 m	47 mph	64%
Bulpetts	79%	1.27 m	61 mph	0%
Scylla	59%	1.86 m	62 mph	0%
Charybdis	90%	3.38 m	64 mph	0%

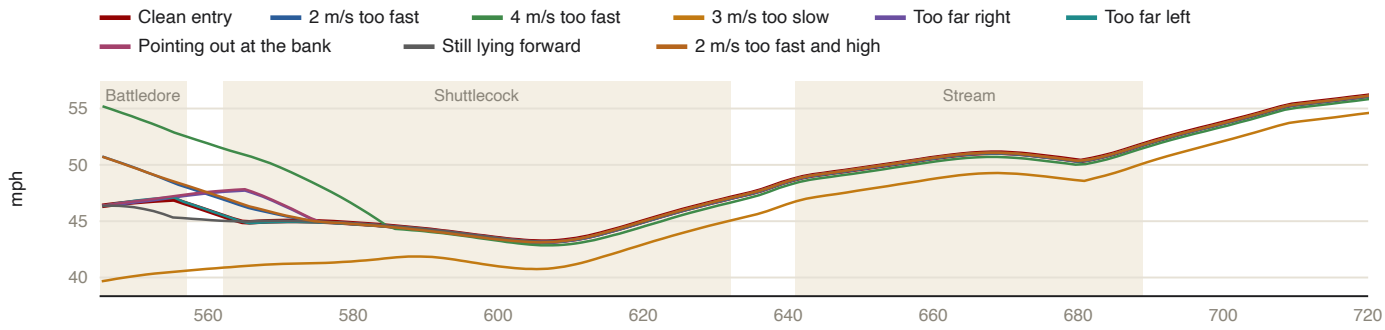
570–590 m	Shuttlecock	Rake (both feet, up to 100%)	49 → 40 mph
656–685 m	Stream	Back on the knives	46 → 46 mph
706–713 m	Bledisloe Straight	Back on the knives	50 → 50 mph
1199–1212 m	Cresta Leap	Back on the knives	69 → 68 mph

Shuttlecock: recovering a bad entry (flat-top)

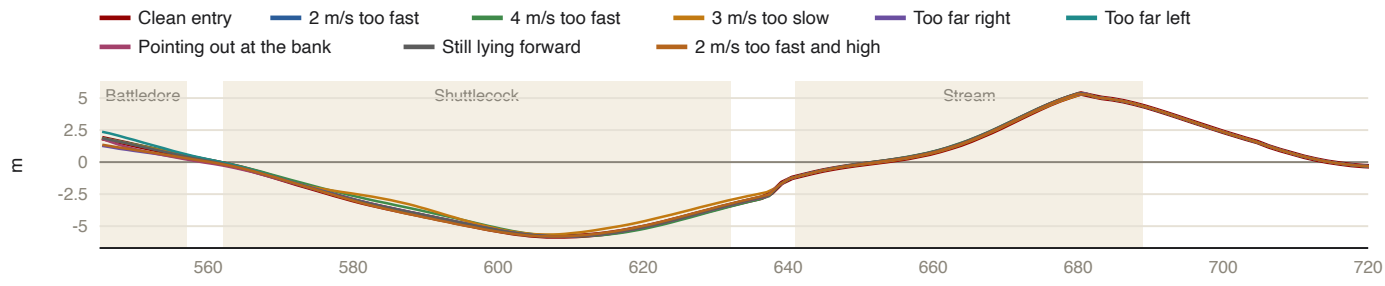
Adelaide on the flat-top, picked up at 545 m (17 m before Shuttlecock's bank begins) in the state the optimum from Junction arrives in, then with one mistake made. For each, the optimiser found the best way on to the finish. “Unchanged” rides the clean optimum's own plan from the bad entry, as a rider who doesn't adapt would. The optimum rides so close to the limit that its plan, picked up mid-run from a state a few centimetres and hundredths of a second off its own, can fall even from the clean entry: on the best line a rider has to keep correcting.

Entry	Speed (mph)	Across (m)	Heading (rad)	Posture	Unchanged plan	Best recovery	Cost vs clean
Clean entry (the optimum)	46	1.90	-0.10	knives	**falls at Shuttlecock**	**25.31 s**	+0.00 s
2 m/s too fast	51	1.90	-0.10	knives	**falls at Shuttlecock**	**25.27 s**	-0.03 s
4 m/s too fast	55	1.90	-0.10	knives	**falls at Shuttlecock**	**25.21 s**	-0.10 s
3 m/s too slow	40	1.90	-0.10	knives	27.44 s	**26.06 s**	+0.75 s
Too far right (towards Shuttlecock’s bank)	46	1.30	-0.10	knives	25.30 s	**25.30 s**	-0.01 s
Too far left (still high on Battledore)	46	2.40	-0.10	knives	**falls at Shuttlecock**	**25.32 s**	+0.02 s
Pointing out at the bank	46	1.90	-0.22	knives	25.29 s	**25.28 s**	-0.03 s
Still lying forward (late onto the knives)	46	1.90	-0.10	forward	**falls at Shuttlecock**	**25.33 s**	+0.02 s
2 m/s too fast and high	51	1.40	-0.10	knives	**falls at Shuttlecock**	**25.26 s**	-0.05 s

SPEED THROUGH SHUTTLECOCK AND STREAM



LINE THROUGH SHUTTLECOCK (THE OUTER BANK IS ON THE RIGHT, –)

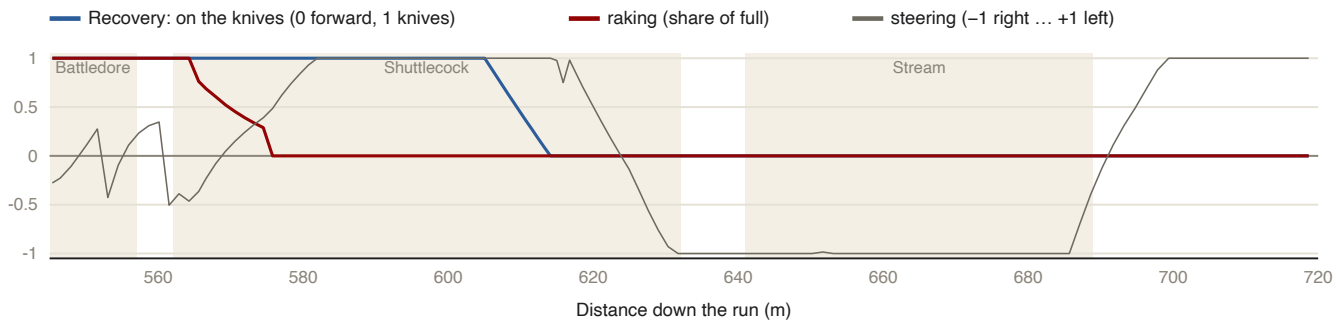


Shuttlecock remedies (flat-top)

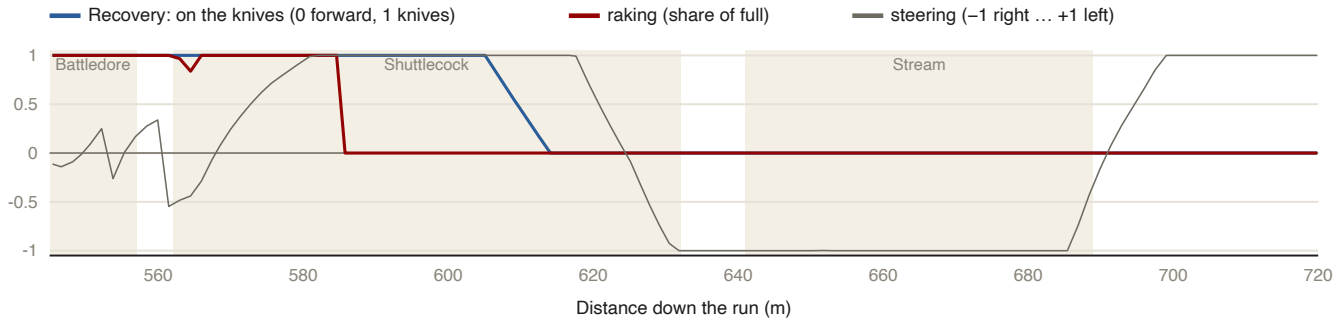
How each best recovery differs from the clean optimum between 545 and 700 m.

Mistake	The remedy
2 m/s too fast	more raking (24 m of full rake against 9); rakes from 545 m (11 m sooner); leaves Shuttlecock at 49 mph (clean 49)
4 m/s too fast	more raking (39 m of full rake against 9); rakes from 545 m (11 m sooner); leaves Shuttlecock at 48 mph (clean 49)
3 m/s too slow	less raking (0 m of full rake against 9); rides lower through the bend (5.65 m out, against 5.80); leaves Shuttlecock at 47 mph (clean 49)
Too far right (towards Shuttlecock's bank)	rakes from 566 m (9 m later); leaves Shuttlecock at 49 mph (clean 49)
Too far left (still high on Battledore)	leaves Shuttlecock at 49 mph (clean 49)
Pointing out at the bank	rakes from 566 m (9 m later); leaves Shuttlecock at 49 mph (clean 49)
Still lying forward (late onto the knives)	on the knives 6 m later (551 m); more raking (13 m of full rake against 9); rakes from 545 m (11 m sooner); leaves Shuttlecock at 49 mph (clean 49)
2 m/s too fast and high	more raking (25 m of full rake against 9); rakes from 545 m (11 m sooner); leaves Shuttlecock at 49 mph (clean 49)

2 M/S TOO FAST: WHAT THE RECOVERY DOES

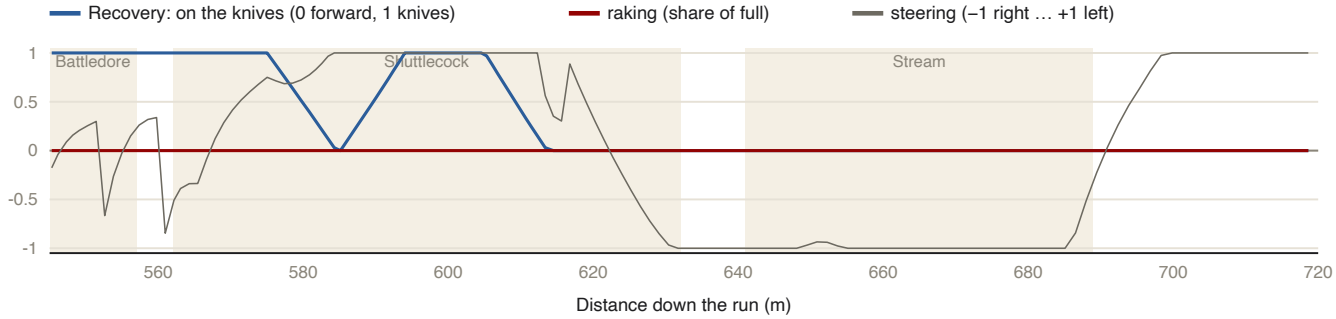


4 M/S TOO FAST: WHAT THE RECOVERY DOES

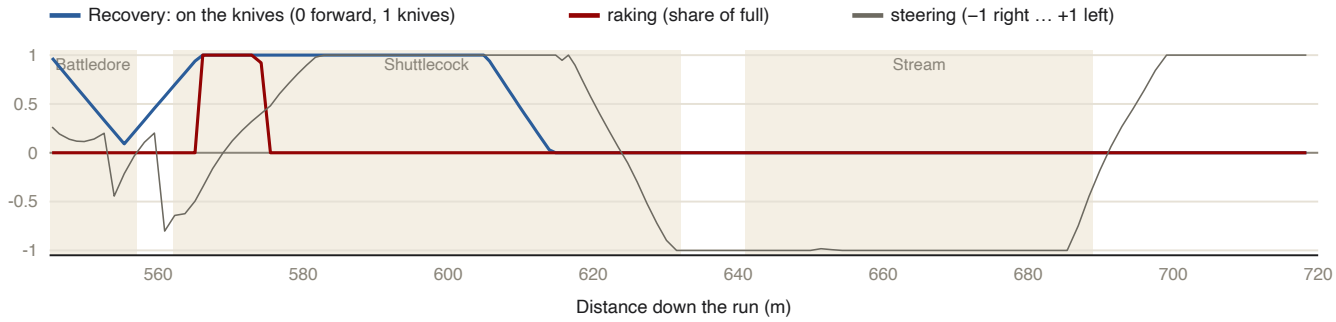


Shuttlecock remedies (flat-top), continued

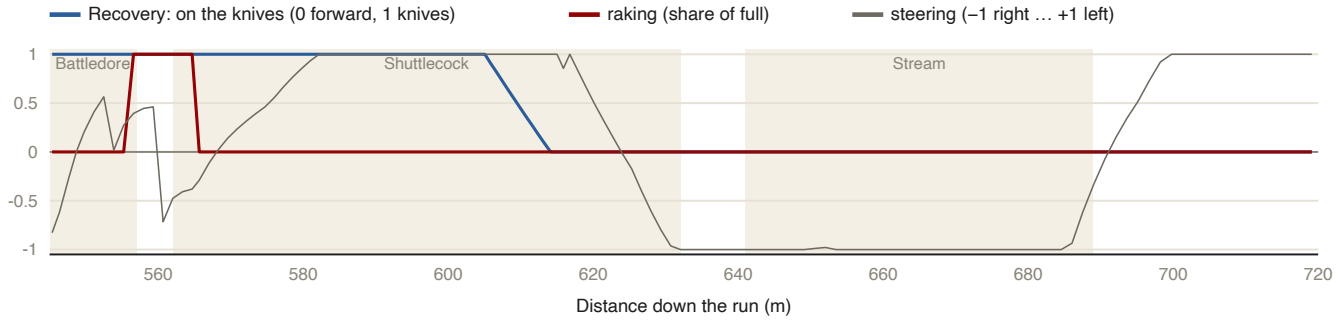
3 M/S TOO SLOW: WHAT THE RECOVERY DOES



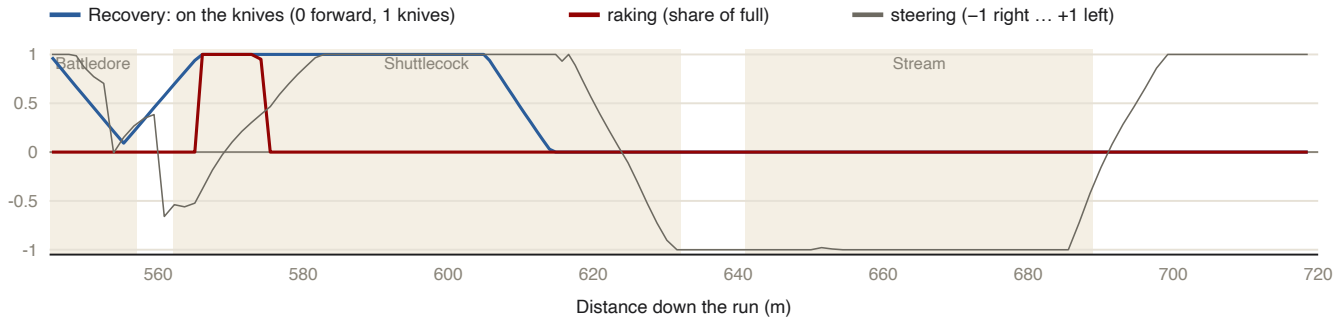
TOO FAR RIGHT (TOWARDS SHUTTLECOCK'S BANK): WHAT THE RECOVERY DOES



TOO FAR LEFT (STILL HIGH ON BATTLEDORE): WHAT THE RECOVERY DOES

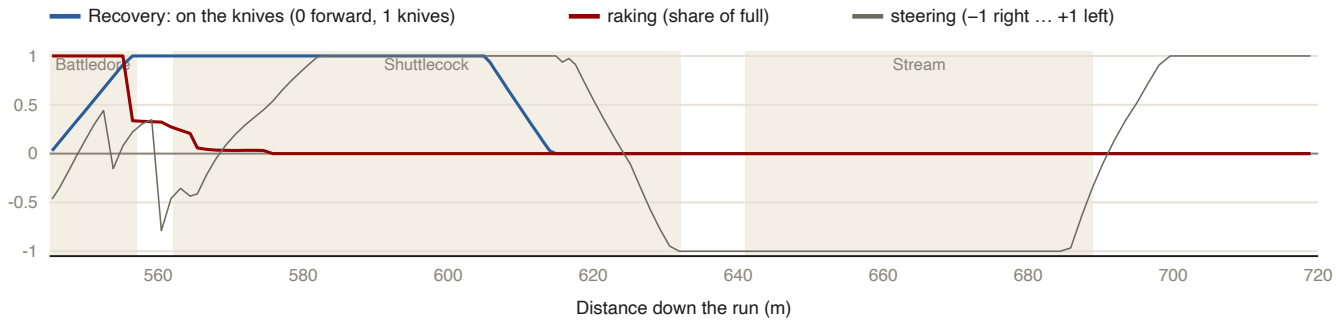


POINTING OUT AT THE BANK: WHAT THE RECOVERY DOES

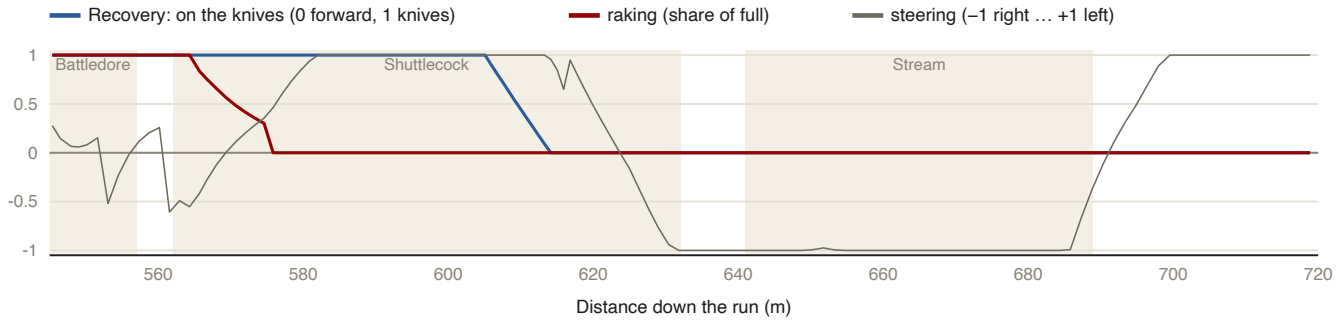


Shuttlecock remedies (flat-top), continued

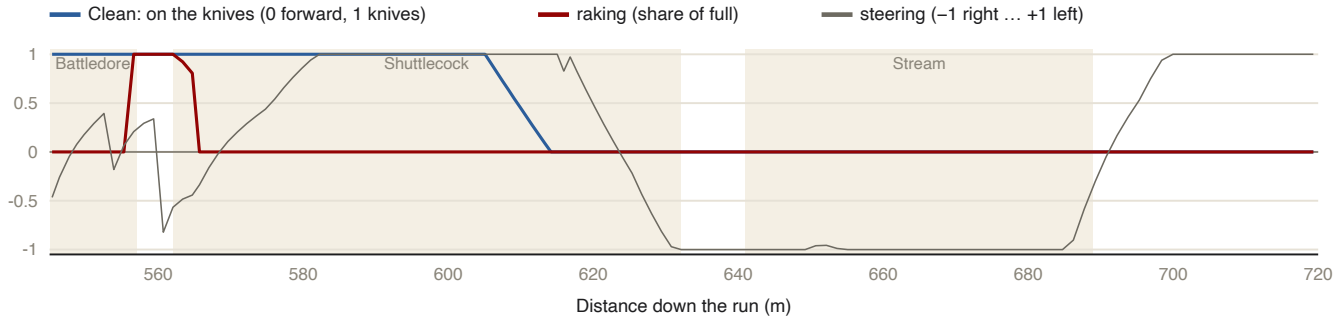
STILL LYING FORWARD (LATE ONTO THE KNIVES): WHAT THE RECOVERY DOES



2 M/S TOO FAST AND HIGH: WHAT THE RECOVERY DOES



CLEAN ENTRY, FOR COMPARISON

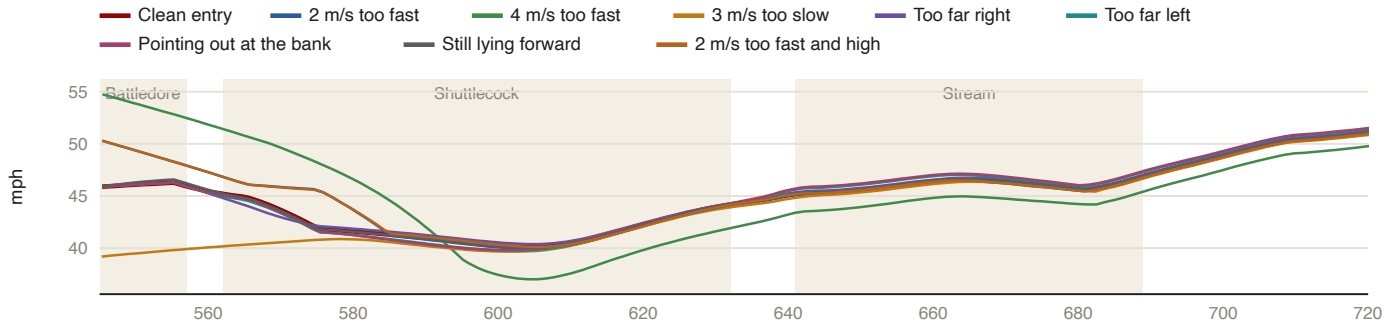


Shuttlecock: recovering a bad entry (traditional)

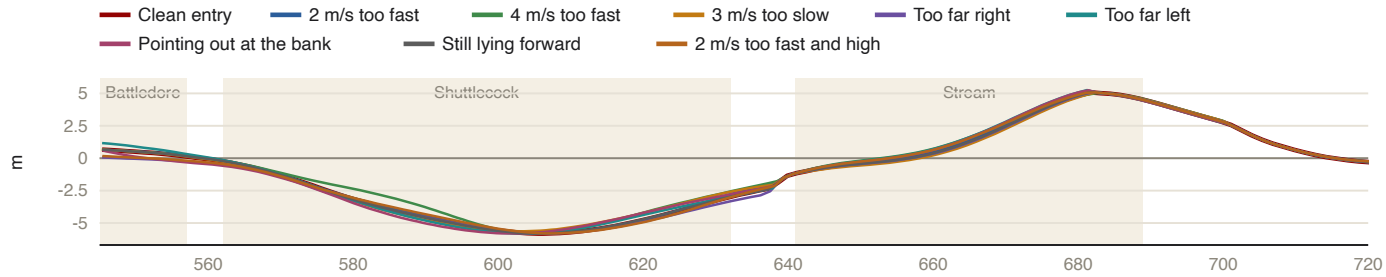
Adelaide on the traditional, picked up at 545 m (17 m before Shuttlecock's bank begins) in the state the optimum from Junction arrives in, then with one mistake made. For each, the optimiser found the best way on to the finish. “Unchanged” rides the clean optimum's own plan from the bad entry, as a rider who doesn't adapt would. The optimum rides so close to the limit that its plan, picked up mid-run from a state a few centimetres and hundredths of a second off its own, can fall even from the clean entry: on the best line a rider has to keep correcting.

Entry	Speed (mph)	Across (m)	Heading (rad)	Posture	Unchanged plan	Best recovery	Cost vs clean
Clean entry (the optimum)	46	0.67	-0.03	knives	27.61 s	**27.50 s**	+0.00 s
2 m/s too fast	50	0.67	-0.03	knives	**falls at Shuttlecock**	**27.47 s**	-0.03 s
4 m/s too fast	55	0.67	-0.03	knives	**falls at Shuttlecock**	**27.77 s**	+0.27 s
3 m/s too slow	39	0.67	-0.03	knives	29.60 s	**27.78 s**	+0.28 s
Too far right (towards Shuttlecock’s bank)	46	0.07	-0.03	knives	27.60 s	**27.45 s**	-0.05 s
Too far left (still high on Battledore)	46	1.17	-0.03	knives	27.65 s	**27.45 s**	-0.05 s
Pointing out at the bank	46	0.67	-0.15	knives	27.63 s	**27.42 s**	-0.07 s
Still lying forward (late onto the knives)	46	0.67	-0.03	forward	**falls at Shuttlecock**	**27.50 s**	+0.00 s
2 m/s too fast and high	50	0.17	-0.03	knives	**falls at Shuttlecock**	**27.44 s**	-0.06 s

SPEED THROUGH SHUTTLECOCK AND STREAM



LINE THROUGH SHUTTLECOCK (THE OUTER BANK IS ON THE RIGHT, –)

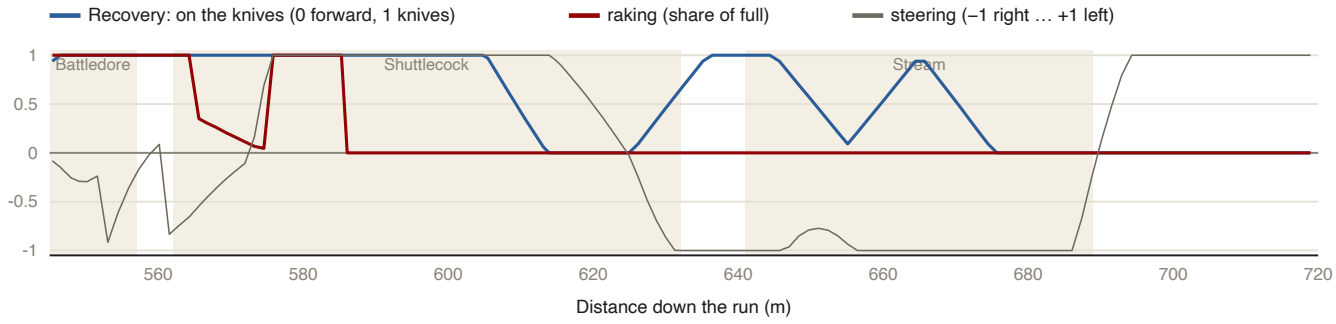


Shuttlecock remedies (traditional)

How each best recovery differs from the clean optimum between 545 and 700 m.

Mistake	The remedy
2 m/s too fast	more raking (32 m of full rake against 16); rakes from 545 m (11 m sooner); leaves Shuttlecock at 45 mph (clean 45)
4 m/s too fast	more raking (52 m of full rake against 16); rakes from 545 m (11 m sooner); leaves Shuttlecock at 43 mph (clean 45)
3 m/s too slow	less raking (0 m of full rake against 16); rides lower through the bend (5.63 m out, against 5.81); leaves Shuttlecock at 45 mph (clean 45)
Too far right (towards Shuttlecock's bank)	leaves Shuttlecock at 45 mph (clean 45)
Too far left (still high on Battledore)	leaves Shuttlecock at 46 mph (clean 45)
Pointing out at the bank	leaves Shuttlecock at 46 mph (clean 45)
Still lying forward (late onto the knives)	on the knives 6 m later (551 m); leaves Shuttlecock at 45 mph (clean 45)
2 m/s too fast and high	more raking (31 m of full rake against 16); rakes from 545 m (11 m sooner); leaves Shuttlecock at 45 mph (clean 45)

2 M/S TOO FAST: WHAT THE RECOVERY DOES

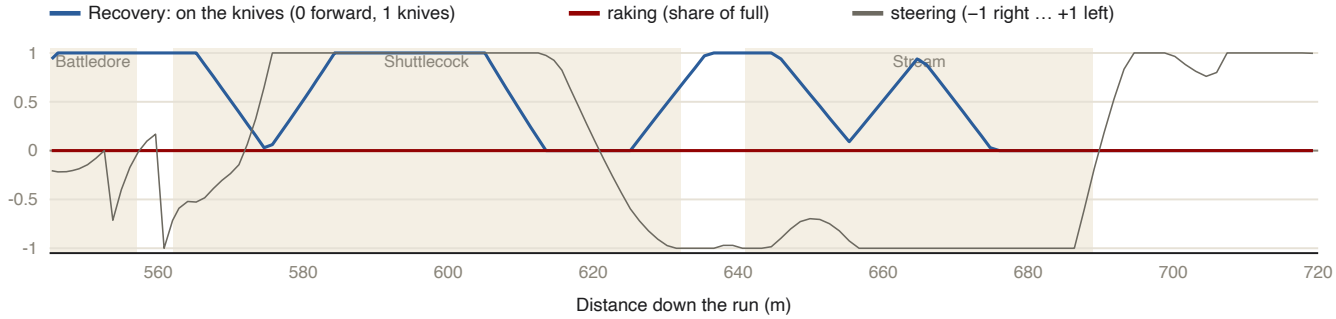


4 M/S TOO FAST: WHAT THE RECOVERY DOES

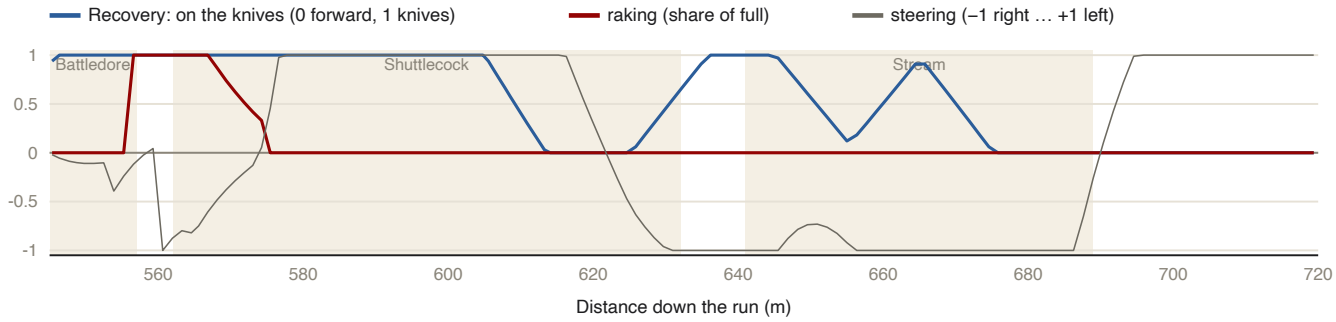


Shuttlecock remedies (traditional), continued

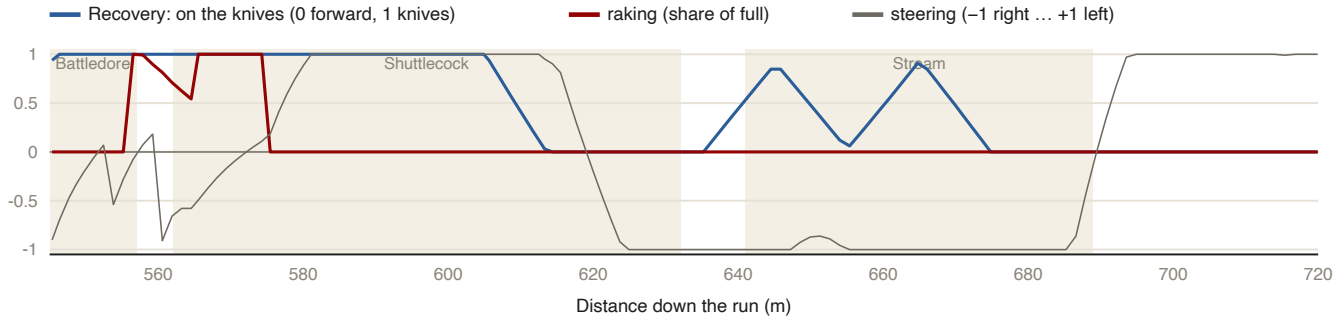
3 M/S TOO SLOW: WHAT THE RECOVERY DOES



TOO FAR RIGHT (TOWARDS SHUTTLECOCK'S BANK): WHAT THE RECOVERY DOES



TOO FAR LEFT (STILL HIGH ON BATTLEDORE): WHAT THE RECOVERY DOES

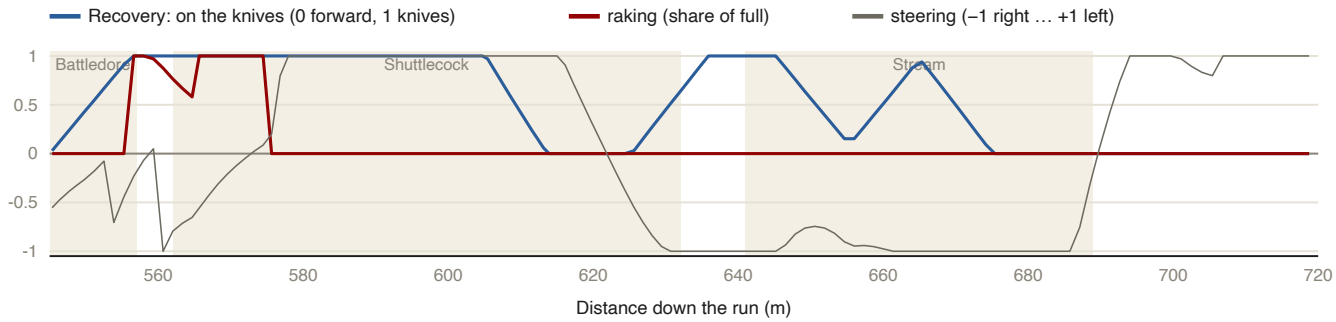


POINTING OUT AT THE BANK: WHAT THE RECOVERY DOES



Shuttlecock remedies (traditional), continued

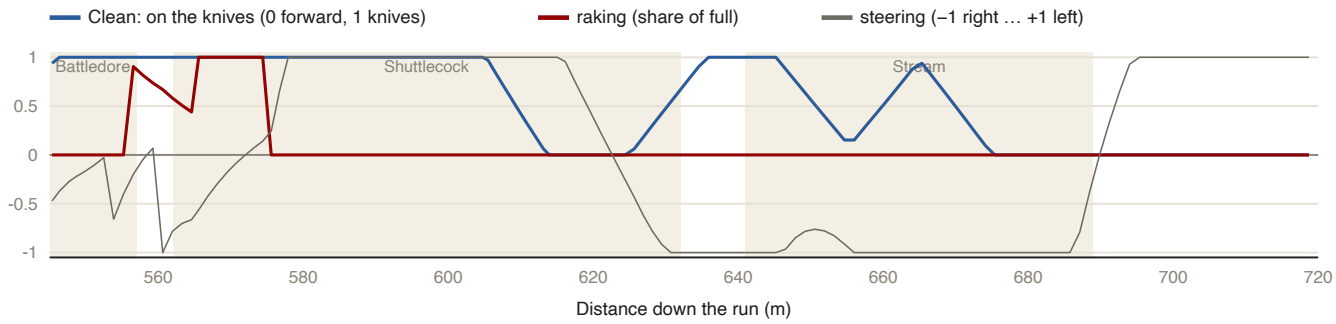
STILL LYING FORWARD (LATE ONTO THE KNIVES): WHAT THE RECOVERY DOES



2 M/S TOO FAST AND HIGH: WHAT THE RECOVERY DOES



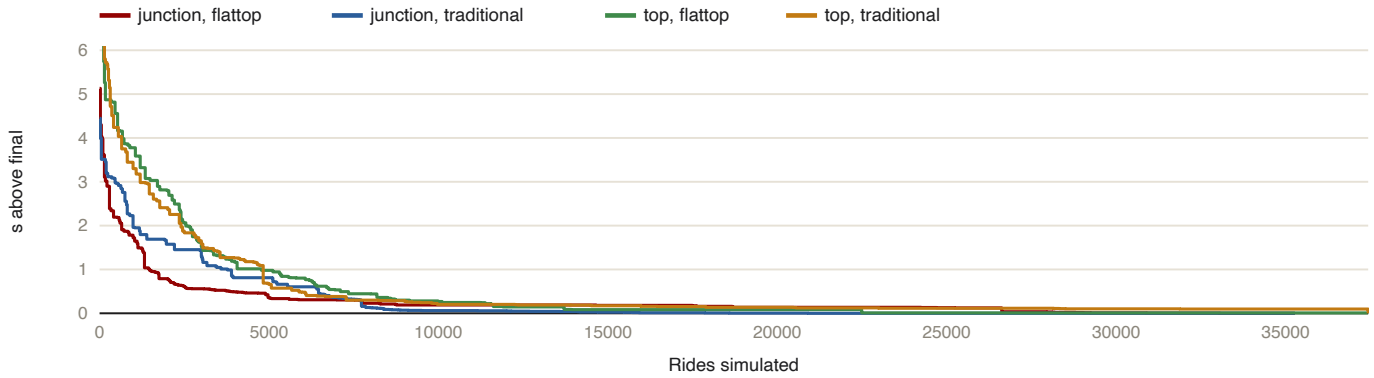
CLEAN ENTRY, FOR COMPARISON



Reading these results

- **These are the game's best, not the real Cresta's.** The optimiser finds whatever the physics allows, including any quirks; a line that looks too good to be true points at the physics rather than at a riding secret.
- **Found, not proven.** CMA-ES finds very good plans, not certainly the best; more rides, or a finer plan (shorter stretches), may find a little more. Repeat runs give the same answer (the search is seeded).
- **One rider per plan.** Each rider and toboggan was optimised on its own: their weight, drag, agility and steering change where to sit back and how hard to rake.
- **Running start** strides every 0.15 s, all seven steps (quicker tapping is worth a little more: about 0.1 s at 0.05 s a stride; the autopilot's 0.28 s costs about 0.5 s). Only the ride after the dive is optimised.
- **An ordinary day.** Except on the page about the day's conditions, every plan is for an ordinary day: no weather, no luck.

HOW THE SEARCH CONVERGED (ADELAIDE; SECONDS ABOVE THE FINAL BEST)



USING THE RESULTS AGAIN

- `optimiser/results/<case>.json`: `genome` (the plan's numbers), `plan` (decoded every 10 m: line, knives, rake cap), `trace` (time, distance, position across the ice, speed, posture, steering, raking, heading and load, 20 times a second), `best` (time, splits, top and finish speed), `baseline` (the autopilot ridden the same way), and for Shuttlecock `unchanged`.
- `optimiser/plan.ts` rides any stored genome again (`ride(track, rider, toboggan, scenario, genome, true)`).
- `npm run optimise` runs everything again (about half an hour; `-- --only junction-flattop` for some), and `npm run optimise:report` rebuilds this report.

Generated 2026-10-29 from 48 optimisation runs.